

AMC 10 Crash Course • Block 1 • Algebra • Lessons 1.1–1.5

Theory, worked examples, and independent sets. Solutions to the independent sets are at the end of the file.

Lesson 1.1. Percents, Ratios, Averages, Motion

THEORY

Percents as multipliers. A 20% increase means multiplying by 1.2; a 25% decrease means multiplying by 0.75. Successive changes multiply: +20% followed by –25% gives $1.2 \cdot 0.75 = 0.9$, that is –10%. Never add percents taken of different bases.

A change of **p%** = multiplication by **(1 + p/100)**. A chain of changes = the product of the multipliers.

Ratios via parts. If two quantities are in the ratio 3 : 4, let k be the size of one part: the quantities are $3k$ and $4k$. Chained ratios are linked through the common quantity: from $a : b = 2 : 3$ and $b : c = 4 : 5$, scale b to 12 to get $a : b : c = 8 : 12 : 15$.

Averages. Mean = sum / count; nearly every average problem is solved by working with the **sum**: find the sum before and after the change. A weighted average lies between the group means, closer to the larger group.

Motion. Distance = speed $\times$ time. Average speed is the TOTAL distance over the TOTAL time, not the average of speeds: over equal distances it equals $\frac{2v_1v_2}{v_1 + v_2}$ and is always less than the arithmetic mean.

WORKED EXAMPLES

EXAMPLE 1 • PERCENTS

A price first rose by 25%, then fell by 20%. How does the final price compare to the original?

→ Multipliers: $1.25 \cdot 0.8 = 1$. The price is unchanged. +25 and –20 look like they should not cancel — but as multipliers they do: $5/4 \cdot 4/5 = 1$. Fractions are faster than decimals here.

EXAMPLE 2 • RATIOS

In a class the ratio of boys to girls is 3 : 4. After 6 more boys join, the ratio becomes 9 : 8. How many students are in the class now?

→ Originally $3k$ and $4k$. Now: $\frac{3k+6}{4k} = \frac{9}{8}$. Cross-multiply: $24k + 48 = 36k$, so $k = 4$. The class now has $(12 + 6) + 16 = \mathbf{34}$ students. Check: $18 : 16 = 9 : 8$. Correct.

EXAMPLE 3 • AVERAGE SPEED

You drive to a town at 60 mph and return along the same road at 40 mph. What is the average speed for the whole trip?

→ Let the one-way distance be d . Time: $d/60 + d/40 = d \cdot 5/120 = d/24$. Average = $2d \div (d/24) = \mathbf{48 \text{ mph}}$. Not 50! The formula $2 \cdot 60 \cdot 40 / (60 + 40)$ gives the same instantly. On the AMC, the arithmetic mean of the speeds almost always appears among the choices as a trap.

EXAMPLE 4 • WEIGHTED AVERAGE

The girls' average score is 90, the boys' is 80, and the class average is 84. What fraction of the class are boys?

→ Let the boys' fraction be m . Then $80m + 90(1 - m) = 84$, so $90 - 10m = 84$ and $m = 0.6$: **60%**. Quick lever check: 84 is 4 away from 80 and 6 away from 90; weights are inverse to distances, $6 : 4 = 3 : 2$ in favor of the boys.

INDEPENDENT SET

1. 20% of 30% of a number equals 12. Find the number.
2. An item's price dropped by 50%. By what percent must it rise to return to the original price?

3. Two liters of a 10% acid solution are mixed with three liters of a 20% solution. What is the concentration of the mixture?
4. $a : b = 2 : 3$ and $b : c = 4 : 5$. Find $a : c$.
5. The average of seven numbers is 15. An eighth number is added, and the average becomes 16. What number was added?
6. A car drives the first half of a route at 30 mph and the second half at 60 mph. What is its average speed?
7. A train 880 feet long passes a pole in 10 seconds. What is its speed in miles per hour?
8. A salary was raised by 10%, and a year later by another 10%. By what percent did it grow over the two years?
9. ★ A container holds a 40% acid solution. A quarter of the contents is removed and replaced with water; then a quarter is removed and replaced with water again. What is the concentration now?

Lesson 1.2. Equations and Systems, Work and Mixtures, Sequences

THEORY

Linear equations and systems. Expand, collect x on the left and numbers on the right, divide. In a two-equation system, solve for the variable whose coefficient is 1 and substitute into the other equation. Plug every answer back in: the check takes ten seconds and catches half of all mistakes.

Work problems. The key is the rate: the fraction of the job done per hour. A pipe that fills a pool in 6 hours does $1/6$ of the job per hour. Rates add: a “6-hour” pipe and a “3-hour” pipe together do $1/6 + 1/3 = 1/2$ of the job per hour, i.e. 2 hours for the whole job.

$$\text{Work per hour} = \frac{1}{\text{time alone}}. \text{ Joint time} = \frac{1}{\text{sum of rates}}.$$

Mixtures. Track the pure substance, not the percents: how many liters of pure juice/acid there were and there are. Added water increases the volume but not the substance.

Sequences. Arithmetic: $a_n = a_1 + (n - 1)d$; $\text{sum} = \frac{a_1 + a_n}{2} \cdot n$ (the mean of the first and last, times the count).

Geometric: $a_n = a_1 r^{n-1}$; $\text{sum} = a_1 \frac{r^n - 1}{r - 1}$ (for $r \neq 1$).

WORKED EXAMPLES

EXAMPLE 1 • SYSTEM

Three pens and two notebooks cost \$24; one pen and four notebooks cost \$18. How much does a pen cost?

→ From the second equation, $p = 18 - 4n$. Substitute: $3(18 - 4n) + 2n = 24$, so $54 - 10n = 24$, $n = 3$ and $p = \$6$. Check: $3 \cdot 6 + 2 \cdot 3 = 24$. Correct.

EXAMPLE 2 • WORK

One pipe fills a pool in 10 hours, another in 15. How long do they take together?

→ Rates: $1/10 + 1/15 = 1/6$ of the pool per hour. Together: **6 hours**. The common trap “ $(10+15)/2$ ” is wrong: together is always faster than the faster pipe alone.

EXAMPLE 3 • MIXTURE

Five liters of juice have concentration 12%. How much water must be added to get 8%?

→ Pure juice: $0.12 \cdot 5 = 0.6$ L, and it does not change. New volume: $0.6 / 0.08 = 7.5$ L. Add **2.5 liters**. The whole problem is one division if you track the substance, not the percents.

EXAMPLE 4 • SEQUENCES

Find the sum $3 + 6 + 12 + \dots + 384$.

→ A geometric sequence: $a_1 = 3$, $r = 2$. The last term is $384 = 3 \cdot 2^7$, so there are 8 terms. Sum $= 3(2^8 - 1)/(2 - 1) = 3 \cdot 255 = 765$. Quick check for $r = 2$: the sum is twice the last term minus the first: $768 - 3 = 765$.

INDEPENDENT SET

1. Solve the equation $5(x - 2) = 3x + 8$.
2. The sum of two numbers is 40, and one is 6 more than the other. Find their product.
3. A painter paints a fence in 4 hours, an apprentice in 12. How long will they take together?
4. Two liters of a 30% solution are mixed with four liters of a 15% solution. What is the concentration of the mixture?
5. In an arithmetic sequence, $a_1 = 7$ and $d = 4$. Find a_{25} .
6. Find the sum of the first 30 odd positive integers: $1 + 3 + 5 + \dots + 59$.
7. In a geometric sequence, $a_3 = 18$ and $a_5 = 162$. Find a_1 .
8. Maya solves 2 problems per minute; Ben solves 3. Ben starts 5 minutes after Maya. How many minutes after Maya starts will they have solved the same number of problems?
9. ★ The sum of the first n terms of a sequence is $S_n = 3n^2 + 2n$. Find the tenth term.

Lesson 1.3. Quadratics: Roots, Vieta's Formulas, the Vertex

THEORY

Roots. Try factoring first: $x^2 - 7x + 12 = (x - 3)(x - 4)$. If it does not factor in ten seconds, use the discriminant: $D = b^2 - 4ac$, roots $(\frac{-b \pm \sqrt{D}}{2a})$. $D > 0$ — two roots, $D = 0$ — one (a double root), $D < 0$ — no real roots.

Vieta's formulas for $x^2 + px + q$: sum of roots $= -p$, product $= q$. The competition payoff: you can learn everything about the roots without ever finding them.

Symmetric expressions via Vieta. $r^2 + s^2 = (r + s)^2 - 2rs$; $r^3 + s^3 = (r + s)^3 - 3rs(r + s)$; $1/r + 1/s = (r + s)/rs$. These three cover most problems.

Vertex. The parabola $ax^2 + bx + c$ has its vertex at $x = -b/2a$ — the minimum point ($a > 0$) or maximum point ($a < 0$). “Find the least value” problems are almost always a vertex substitution. And the sign: for $a > 0$, the quadratic is negative strictly between its roots and positive outside them — that is all you need for quadratic inequalities.

WORKED EXAMPLES

EXAMPLE 1 • FACTORING

Solve $x^2 - 7x + 12 = 0$.

→ Look for two numbers with sum 7 and product 12: these are 3 and 4. Roots: **3 and 4**. No discriminant needed: factoring is faster and avoids arithmetic slips.

EXAMPLE 2 • VIETA

Let r and s be the roots of $x^2 - 5x + 3 = 0$. What is $r^2 + s^2$?

→ By Vieta's formulas, $r + s = 5$, $rs = 3$. Then $r^2 + s^2 = 25 - 2 \cdot 3 = \mathbf{19}$. The roots themselves are irrational, and finding them wastes two minutes.

EXAMPLE 3 • VERTEX

Find the least value of $x^2 - 6x + 11$.

→ Vertex at $x = 6/2 = 3$. Value: $9 - 18 + 11 = \mathbf{2}$. Alternatively, complete the square: $(x - 3)^2 + 2 \geq 2$, which shows the minimum is 2, at $x = 3$.

EXAMPLE 4 • PARAMETER

For which k does $x^2 + kx + 9 = 0$ have exactly one root?

→ Exactly one root $\Leftrightarrow D = 0$: $k^2 - 36 = 0$, so $k = \mathbf{6 \text{ or } -6}$. The classic way to lose a point here is forgetting the negative value.

INDEPENDENT SET

1. Solve $x^2 = 5x$. Give all roots.
2. Find the largest root of $x^2 - 2x - 15 = 0$.
3. Roots r and s satisfy $r + s = 6$ and $rs = 4$. What is $1/r + 1/s$?
4. Let r and s be the roots of $x^2 - 4x + 1 = 0$. What is $r^3 + s^3$?
5. Find the greatest value of $-x^2 + 8x - 7$.
6. The number 2 is a root of $x^2 - 6x + c = 0$. Find the other root.
7. How many integers satisfy $x^2 - 5x + 6 < 0$?
8. A quadratic with integer coefficients has roots $3 + \sqrt{2}$ and $3 - \sqrt{2}$. What is the product of the roots?
9. ★ Let r and s be the roots of $x^2 - 3x - 2 = 0$. A quadratic of the form $x^2 + px + q = 0$ has roots $r + 1$ and $s + 1$. What is q ?

Lesson 1.4. Identities and Tricks: Squares, Cubes, Telescoping**THEORY**

Three identities to know on sight. $(a \pm b)^2 = a^2 \pm 2ab + b^2$; difference of squares $a^2 - b^2 = (a - b)(a + b)$; sum and difference of cubes $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$. The difference of squares is the biggest computational shortcut: $101^2 - 99^2 = 200 \cdot 2$.

If $s = x + 1/x$, then $x^2 + 1/x^2 = s^2 - 2$, and $x^3 + 1/x^3 = s^3 - 3s$. Raise what you know to a power and subtract the extra.

Telescoping. A long sum or product where neighboring terms cancel: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$ — everything inside collapses, leaving the first and last pieces. If you see a sum of a hundred fractions, look for telescoping.

Grouping. $ab + a + b + 1 = (a + 1)(b + 1)$. Adding one and factoring is a standard move in integer problems.

WORKED EXAMPLES

EXAMPLE 1 • DIFFERENCE OF SQUARES

Compute $101^2 - 99^2$ without long multiplication.

→ $(101 - 99)(101 + 99) = 2 \cdot 200 = \mathbf{400}$. On the AMC such computations are embedded in bigger problems: recognize the difference of squares instantly.

EXAMPLE 2 • $X + 1/X$

Given $x + 1/x = 3$, find $x^2 + 1/x^2$ and $x^4 + 1/x^4$.

→ Square it: $9 = x^2 + 2 + 1/x^2$, so $x^2 + 1/x^2 = 7$. Same move again: $49 = x^4 + 2 + 1/x^4$, so $\mathbf{47}$. Each squaring brings an extra 2 — subtract it.

EXAMPLE 3 • TELESCOPING

Compute the sum $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{99 \cdot 100}$.

→ Each term: $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. The sum collapses: $1 - 1/100 = \mathbf{99/100}$.

EXAMPLE 4 • GROUPING

Positive integers a and b satisfy $ab + a + b = 63$. What is the least possible value of $a + b$?

→ Add 1: $(a + 1)(b + 1) = 64$. Ways to write 64 as a product of two factors ≥ 2 : $2 \cdot 32$, $4 \cdot 16$, $8 \cdot 8$. The sums $a + b$: 32, 18, 14. The least is $\mathbf{14}$ (at $a = b = 7$): with a fixed product, the sum is smallest when the factors are closest together.

INDEPENDENT SET

1. Compute $51^2 - 49^2$.
2. Compute $(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})$.
3. Given $x + 1/x = 6$, find $x^3 + 1/x^3$.
4. Compute $999 \cdot 1001$ without long multiplication.
5. Compute the product $(1 - \frac{1}{2})(1 - \frac{1}{3}) \dots (1 - \frac{1}{100})$.
6. Numbers a and b satisfy $a + b = 7$ and $ab = 10$. Find $a^3 + b^3$.
7. Compute the sum $\frac{1}{\sqrt{2} + 1} + \frac{1}{\sqrt{3} + \sqrt{2}} + \dots + \frac{1}{\sqrt{25} + \sqrt{24}}$.
8. Compute $\frac{2^{10} - 1}{2^5 - 1}$.
9. ★ Compute $(2 + 1)(2^2 + 1)(2^4 + 1)(2^8 + 1) + 1$.

Lesson 1.5. Absolute Value: Equations, Inequalities, Distances

THEORY

Absolute value is distance. $|x|$ is the distance from x to zero; $|x - a|$ is the distance from x to the point a . Half of all absolute-value problems are solved faster by a number-line picture than by case analysis.

$$|f| = c \text{ (for } c > 0) \Leftrightarrow f = c \text{ or } f = -c. \text{ For } a > 0: |x| < a \Leftrightarrow -a < x < a; |x| > a \Leftrightarrow x < -a \text{ or } x > a.$$

Sum of distances. $|x - a| + |x - b|$ is the total distance from x to two points. Between the points it is constant and equals $|a - b|$; outside, it grows. For an odd number of points, the minimum of the sum of distances is at the middle (median) point.

Nested absolute values. $||x| - 4| = 2$ unwraps layer by layer, outside in: first $|x| - 4 = \pm 2$, then each case separately. Care beats speed: count the roots at the end.

WORKED EXAMPLES

EXAMPLE 1 • EQUATION

Solve $|2x - 3| = 7$.

→ $2x - 3 = 7$ or $2x - 3 = -7$: roots **5 and -2**. Check by substitution: $|7| = 7$, $|-7| = 7$. Correct.

EXAMPLE 2 • SUM OF DISTANCES

What is the least value of $|x - 1| + |x - 5|$, and for which x is it attained?

→ This is the total distance from x to the points 1 and 5. Between them it is constant and equals the distance between the points: **4**, for every x in $[1, 5]$. No case analysis: one picture.

EXAMPLE 3 • INEQUALITY

How many integers satisfy $|x - 2| < 3$?

→ $-3 < x - 2 < 3$, that is $-1 < x < 5$. Integers: 0, 1, 2, 3, 4 — **five**. A frequent mistake is counting the endpoints: with a strict inequality, -1 and 5 are not included.

EXAMPLE 4 • NESTED ABSOLUTE VALUES

How many solutions does $||x| - 4| = 2$ have?

→ First layer: $|x| - 4 = 2$ or -2 , so $|x| = 6$ or $|x| = 2$. Second layer: $x = \pm 6$ and $x = \pm 2$. In total, **4 solutions**.

INDEPENDENT SET

1. Solve $|x + 4| = 9$ and find the sum of the roots.
2. How many solutions does $|3x - 6| = 0$ have?
3. How many integers satisfy $|x - 3| \leq 5$?
4. How many integer solutions does $|x + 2| + |x - 4| = 6$ have?
5. Solve $|x - 1| = |x - 7|$.
6. What is the least value of $|x - 2| + |x - 6| + |x - 9|$?
7. Find the least positive integer solution of $|5 - 2x| > 3$.
8. How many solutions does $||x - 1| - 3| = 2$ have?
9. ★ How many solutions does $|x - 1| + |x - 3| = x$ have?

Solutions to the independent sets

LESSON 1.1

1. $0.06x = 12$, so $x = \mathbf{200}$.
 2. $0.5 \cdot t = 1$, so $t = 2$: a rise of **100%**. Percents down and up are not symmetric, because the bases differ.
 3. Pure acid: $0.2 + 0.6 = 0.8$ L out of 5 L: **16%**. This is the weighted average of 10 and 20 with weights 2 : 3.
 4. $b = 12$: then $a = 8$, $c = 15$. Answer: **8 : 15**.
 5. Before: 105; after: $16 \cdot 8 = 128$. The number added is **23**.
 6. $2 \cdot 30 \cdot 60 / (30 + 60) = \mathbf{40 \text{ mph}}$.
 7. 88 ft/s. Since $60 \text{ mph} = 88 \text{ ft/s}$, the speed is **60 mph**.
 8. $1.1 \cdot 1.1 = 1.21$: by **21%**. The extra 1% is “percent on percent”: ten percent of ten.
- ★ Each replacement multiplies the amount of acid by $3/4$: $40\% \cdot (3/4)^2 = \mathbf{22.5\%}$.

LESSON 1.2

1. $5x - 10 = 3x + 8$, so $2x = 18$, $x = \mathbf{9}$.
 2. $2x + 6 = 40$, $x = 17$: the numbers are 17 and 23, product **391**.
 3. $1/4 + 1/12 = 1/3$ of the fence per hour, so **3 hours**.
 4. 1.2 L of substance out of 6 L: **20%**.
 5. $7 + 24 \cdot 4 = \mathbf{103}$.
 6. $(1 + 59)/2 \cdot 30 = \mathbf{900}$. A nice fact: the sum of the first n odd positive integers is always n^2 .
 7. $r^2 = 162/18 = 9$. Then $a_1 = 18/r^2 = \mathbf{2}$ (the sign of r does not matter: we divide by r^2).
 8. Maya has worked t minutes, Ben $t - 5$: the equation $2t = 3(t - 5)$ gives $t = \mathbf{15 \text{ minutes}}$. Check: both have solved 30 problems.
- ★ $S_{10} = 320$, $S_9 = 261$: the tenth term is **59**. The trick works for any sum formula: a term is the difference of consecutive sums.

LESSON 1.3

1. $x(x - 5) = 0$: roots **0 and 5**. Dividing by x loses the root 0.
 2. $(x - 5)(x + 3) = 0$: roots 5 and -3 . The largest is **5**.
 3. $(r + s)/rs = 6/4 = \mathbf{3/2}$.
 4. $64 - 3 \cdot 1 \cdot 4 = \mathbf{52}$.
 5. Vertex at $x = 4$: $-16 + 32 - 7 = \mathbf{9}$.
 6. The sum of the roots is 6, so the other root is **4** (and incidentally $c = 8$).
 7. $(x - 2)(x - 3) < 0$ for $2 < x < 3$. There are **no integers there: 0**.
 8. $9 - 2 = 7$. The equation itself: $x^2 - 6x + 7 = 0$.
- ★ $(r + 1)(s + 1) = rs + r + s + 1 = -2 + 3 + 1 = \mathbf{2}$. No need to find the roots: Vieta's formulas give everything.

LESSON 1.4

1. $2 \cdot 100 = \mathbf{200}$.
2. $7 - 3 = \mathbf{4}$.

3. $216 - 18 = \mathbf{198}$.

4. $10^6 - 1 = \mathbf{999,999}$.

5. $\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{99}{100}$: everything cancels down to $\mathbf{1/100}$.

6. $343 - 210 = \mathbf{133}$.

7. Each term equals $\sqrt{(k+1)} - \sqrt{k}$; the telescope gives $\sqrt{25} - 1 = \mathbf{4}$.

8. $(2^5 - 1)(2^5 + 1)/(2^5 - 1) = 2^5 + 1 = \mathbf{33}$.

★ $(2-1)(2+1)(2^2+1)(2^4+1)(2^8+1) = 2^{16} - 1$, so the expression equals $2^{16} = \mathbf{65,536}$.

LESSON 1.5

1. The solutions are 5 and -13 ; they are symmetric about -4 , so the sum is $\mathbf{-8}$.

2. Only $3x - 6 = 0$: **one solution**, $x = 2$.

3. From -2 to 8 inclusive: **11 integers**.

4. The distance between -2 and 4 is 6 , so the equation holds on the whole segment $[-2, 4]$. Integer solutions: **7**.

5. The midpoint: $x = \mathbf{4}$.

6. At $x = 6$: $4 + 0 + 3 = \mathbf{7}$.

7. Positive integers with $x > 4$ start at **5** (1 through 4 fail: substitute and check).

8. $|x - 1| = 5$ gives 6 and -4 ; $|x - 1| = 1$ gives 2 and 0 . In total, **4 solutions**.

★ For $x < 1$: $4 - 2x = x$, $x = 4/3$ — not in the interval. For $1 \leq x \leq 3$: $2 = x$ — works. For $x > 3$: $2x - 4 = x$, $x = 4$ — works. In total, **2 solutions** (2 and 4).